

INDUSTRIAL AUTOMATION

The Industrial Automation track develops graduates into highly skilled automation engineers capable of designing, integrating, and optimizing smart industrial systems. Across 1,506 hours, participants gain practical expertise in PLCs, HMI/SCADA, industrial robotics, motion control, process automation, industrial communication networks, and Industry 4.0 solutions—preparing them for high-demand careers across Egypt's manufacturing, energy, utilities, and smart factory sectors.

JOB PROFILES

**PLC & SCADA Engineer · Process Control Engineer ·
DCS Application Engineer · Instrumentation & Control Engineer**



TOTAL HOURS

1,301 hrs

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Good

Minimum grade required



Where graduates work

Schneider Electric

Siemens

ABB

Giza Systems

Advanced Intillegent Solutions

Systech

Controlware

Area Technology

Mas Automation

Yokogawa Automtion



Who may apply

Open to graduates in electrical engineering, mechatronics engineering, and electromechanical with a minimum grade of 'good' or higher.



Selection process

01 IQ & English Exam

💡 Online exam covering problem-solving aptitude and English proficiency.

02 Technical Exam

💡 Computer-based assessment covering the program's core technical fundamentals.

03 Technical Interview

💡 Interview with the technical committee.

04 Interpersonal Interview

💡 Candidates who pass the technical interview are invited to this final stage.

Program content – 1,301 hours

CATEGORY	HOURS	SHARE	KEY COURSES
● Core Courses	560 hrs	43%	Industrial Process Control , Advanced Process Control, Home Automation using KNX, Instrumentation for Automation , Electrical Power Engineering Principles for Automation, PLC Applications using Siemens , Industrial Data Communications and Networking, Distributed Control Systems, Radio Telemetry Systems for Industry and RTUs, SCADA and HMI Systems for Industry, Functional Safety and Hazard Operability, Motors and Drives, Advanced Pneumatic and Hydraulic Control Systems, TurboMachinery Control using Triconix, Industrial Control System Security Essentials, Robotics and Computer Vision
● Fundamental Courses	350 hrs	27%	Introduction to Programming using C, Object-Oriented Programming Concepts, Introduction to Database Fundamentals , Data Structures and Algorithms, Introduction to Automation, Introduction to PLCs, Panel Design using AutoCAD Electric, Introduction to Generative AI and prompt engineering, Modeling and Simulation -- Introduction to MATLAB, Lean Six Sigma
● Soft Skills	91 hrs	7%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor
● Graduation Project	300 hrs	23%	

Hands-on labs are conducted in-person. Online components cover theory and self-study modules on MaharaTech.

Student Deliverable

Every graduate must obtain an international certification relevant to their specialisation — Tia portal - Ecostructure SW issued by Schneider Electric or Siemens, confirming professional readiness for the field.

Before you apply

Explore these resources before your interview. Applicants are expected to have reviewed this material.

- Automation Systems
<https://www.udemy.com/course/sas-substationautomation-withhossamembaby/>
- Mechatronics and Industrial Internet of Things
<https://www.udemy.com/course/mechatronics-and-industrial-internet-of-things/>
- Electrical Design in Industrial Automation - The Insight
<https://www.udemy.com/course/electrical-design-the-insight/>
- TIA Portal for Beginners
<https://www.udemy.com/course/tia-portal-for-beginners/>

APPLY AT

ITI.GOV.EG

Applications open each intake period. Check the website for current dates and deadlines.